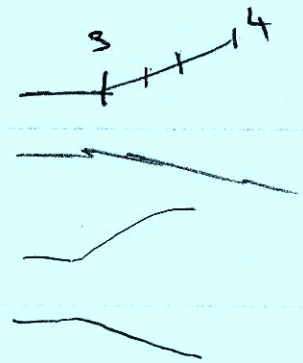
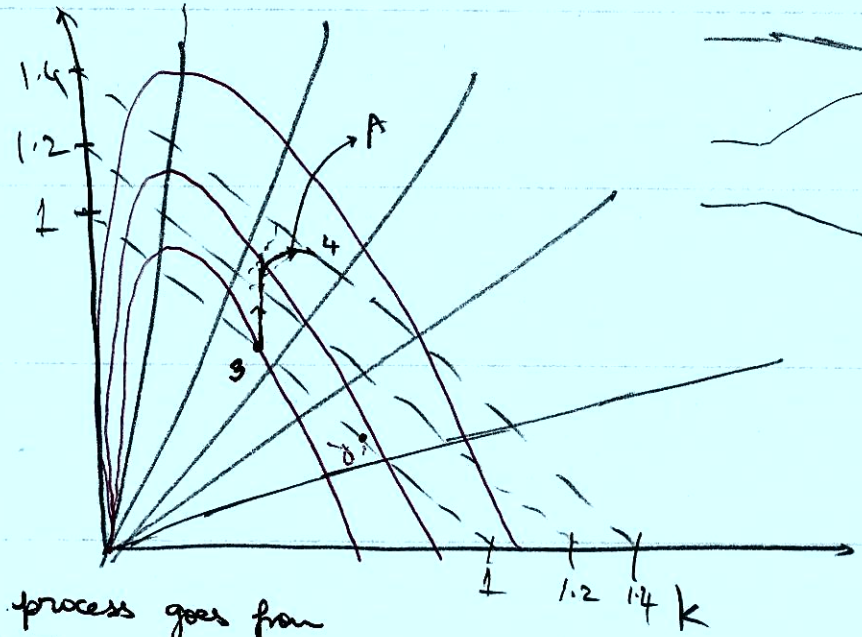
□ H-k diag. of shock-free (no isol. shocks) scramjet op.



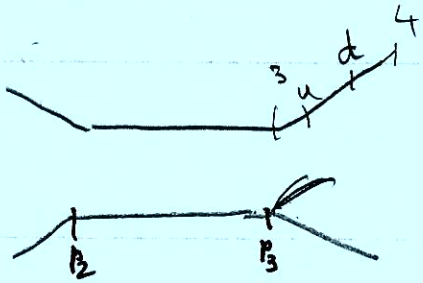
□ At burner entry, heat add^n rate is max $\Rightarrow \frac{dT_t}{dx}$ is greatest, the process path is close to a const-p process, as the change in pr. due to comb. is counteracted by Area increase



□ Further d/s towards burner exit, $\frac{dT_t}{dx}$ tapers off, Area relief dominates $\Rightarrow$ static pr $\downarrow$ ($k\uparrow$) $\Rightarrow$ process goes from const p $\rightarrow$ const M $\rightarrow$ const T $\rightarrow$ const T_t

□ This is called a "shock-free operation"

- Fuel inj @ ③; heat add^n begins @ u; pr rise between u–d
- 's' where the –ve pr. grad begins, coincides w/ 'd'
- No flow sep.
- No shocks before '3'
- $p_3 = p_2$



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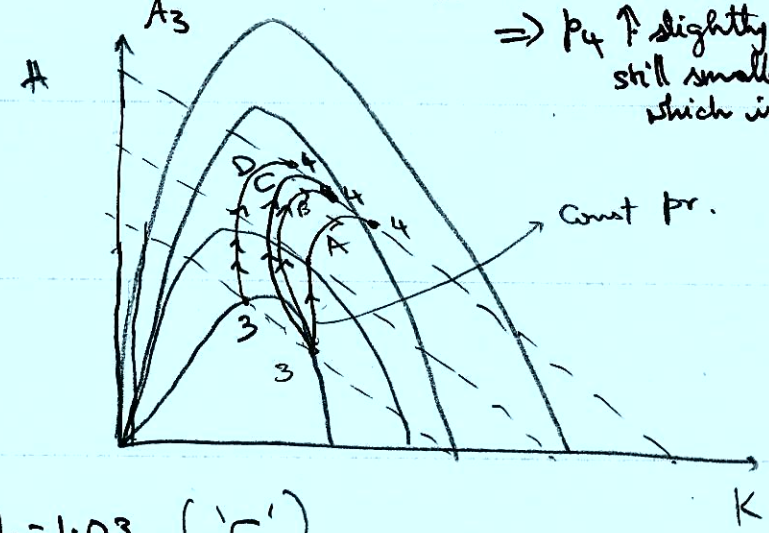
- For the above e.g., to create a const pr. w/ $M_3 = \underline{\underline{2}}$; $\frac{A_4}{A_3} = 2 \Rightarrow M_s = 1.33$. ('A')

  check the text.

- To increase the efficiency, we can reduce $\frac{A_4}{A_3} = 1.73 \Rightarrow M_s = 1.19 \Rightarrow$ No flow sep ('B')
  - Still shock-free operation

  $\Rightarrow p_4 \uparrow$ slightly, but still smaller than p_3 which is the target



- Let's reduce $\frac{A_4}{A_3}$ to 1.57 $\Rightarrow M_s = 1.03$ ('C')

  $\rightarrow$ We will revisit this for learning abt the flow pattern

- $\frac{A_4}{A_3} = 1.55 \Rightarrow$ choked. $\Rightarrow$ $M_3 < 1$ in the burner, $M_4 > 1$ out of the burner.

  $\because M_3 < 1 \Rightarrow$ ram operation

  $\Rightarrow$ We have transitioned from scram to ram mode by heat add^n.

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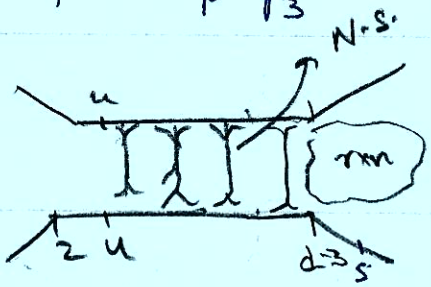
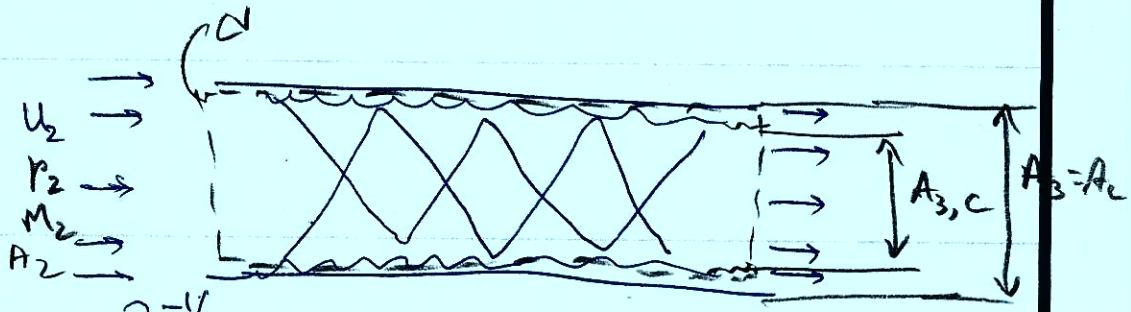
□ Ram-mode.

- If the heat release is adequately large, the O.S. train transitions to a N.S. train & this N.S. train lives entirely w/in the isolator
- Subsonic flow along the core ⇒ back pr @ the burner entry uniformly distributed across the entire cross-section.
- $\because M_3 < 1 \Rightarrow$ Ram-mode.
- Pr rise begins @ 'u' & reaches max @ 'd', where $p = p_3$
- Note 'd' & '3' coincide in ram-mode.
  - Fuel burns very close to station '3' since subsonic mixing is very fast.
- '3'–'s' is very short



□ C-V. analysis of isolator

- Neglect wall friction



$$- M_3 = \left\{ \frac{\gamma_b^2 M_2^2 \left(1 + \frac{\gamma_b - 1}{2} M_2^2\right)}{\left(1 + \gamma_b M_2^2 - \frac{p_3}{p_2}\right)^2} - \frac{\gamma_b - 1}{2} \right\}^{-1/2}$$

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$$\frac{p_3}{p_2} = 1 + \gamma_b M_2^2 - \gamma_b M_2 M_3 \sqrt{\frac{1 + \frac{\gamma_b - 1}{2} M_2^2}{1 + \frac{\gamma_b - 1}{2} M_3^2}}$$

$$\frac{A_{3c}}{A_2} = \frac{1}{\gamma_b M_3^2}\left[\frac{p_2}{p_3}\left(1 + \gamma_b M_3^2\right) - 1\right]$$

□ 1-D flow analysis of burner.

$$\tau(x) = 1 + (\tau_b - 1)\left\{\frac{\theta \chi}{1 - (\theta - 1)\chi}\right\} \qquad \theta \geq 1$$

$$\tau(x) \equiv \frac{T_t(x)}{T_{t2}} \; ; \; \chi \equiv \frac{x - x_i}{x_4 - x_i} \; ; \; x_i : \text{axial loc where heat addn begins}$$

Scramjets : θ : 1-10 depending on mode of fuel/air mixing.

For ramjets θ : 40-50 ; $x_i = x_3$

$$T(x) = T_2 \frac{T_t(x)}{T_{t2}} \cdot \left[\frac{1 + \frac{\gamma_b - 1}{2} M_2^2}{1 + \frac{\gamma_b - 1}{2} M(x)}\right]$$

$$p(x) = p_2 \cdot \frac{A_2}{A_c(x)} \cdot \frac{M_2}{M_2(x)} \sqrt{\frac{T(x)}{T_2}}$$

$$K(x) = \frac{\gamma_b - 1}{2} \cdot M^2(x) \cdot H(x) \; ; \qquad H(x) = \frac{T(x)}{T_{t2}}$$

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